

Class Exercise 8/9: Method

Instructions: **Organize** the information needed to write the Method section of your paper.

Requirements: To successfully complete this exercise, you are required to:

1. **follow along** with your instructor to complete this exercise
2. **fill in** the necessary information for each subsection of your Methods section
 - Participants
 - Materials and Procedure

Please use a **word processing program** (such as MS Word) to complete this exercise.

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General Description

This in-class exercise is designed to help you keep track of the information needed to write the Methods section of your paper (Participants, Materials, Procedure).

We are conducting a 2(IH condition: intellectual humility vs. control) x 2 (PTR condition: forgiving vs. unforgiving) between-subjects ANOVA. We aim to find a similar effect of intellectual humility on transgressors' apology behavior as found in Ludwig and colleagues (2022) using an experimental design, and to extend the original study by investigating the effects of the victim's post-transgression response.

For readers to understand the purpose of our class experiment (see box above), you must clearly define what you mean when using specific terms to describe our variables. Be explicit about how each variable was manipulated or measured (i.e., its operationalization) while also describing the order in which participants completed the study (Materials and Procedure subsection), and who the participants were (Participant subsection).

The “Who” of data collection: participant demographics (Participant subsection)

Jot down some thoughts below regarding the important participant information you should include in your Participants subsection of the Methods.

- Age
- Racial demographics
- Gender demographics
- Type of sample
- Total sample size and final sample size

In the next class exercise, we will work to create APA-style tables summarizing participant demographic data.

The “What” and “How” of our experiment: Describing operationalization/manipulations and the step-by-step process to complete the study

First, as a comprehension check, please identify the IVs and DVs below, as well as other information we collected (i.e., participant data, other measures, etc.; see below)

- IV 1: IH condition
levels: Intellectual humility vs. neutral condition
- IV 2: Victim’s post-transgression response
levels: Unforgiving vs. forgiving condition
- DV: Apology quality
- Additional variables collected Apology behavior (avoidance, defensiveness, accountability)
- Demographic variables: Race/ethnicity, gender, age, and highest level of education

Now, describe your variables in the order that participants completed the study. Here, you want to thoroughly describe how each IV was manipulated and how each DV was measured (aka your specific operational definitions!). *Think of this subsection as the ingredient list and recipe when baking or cooking, which is combined into one subsection.*

Another perspective: for the Materials & Procedures subsection, think about how your description of your manipulations and measures will convince your reader that your materials are appropriate to effectively test your research questions

1. How did you operationalize your first independent variable and its conditions?

For the IH condition, participants had to think of a time they acted with intellectual humility. The aim of this is to temporarily increase their intellectual humility. The neutral condition acts as a control, so there is no attempt to raise their intellectual humility.

2. How did you operationalize your second independent variable and its conditions?

Once participants complete the first question, they are then instructed to imagine themselves in a situation. There are two conditions, forgiving and unforgiving. This situation is the same for both conditions, but the victim's response in each situation is changed. In the forgiving condition, the victim's post transgression response is forgiving of the transgressor. In unforgiving conditions, the victim's post-transgression response is not forgiving. This allows us to see if the way a victim's responds has any impact on the apology behavior of the transgressor.

3. How did you operationalize your primary dependent variable? (note: also describe other measures, including defensiveness and avoidance)

Participants were tasked with ranking the likeliness of how they would respond, given a specific response (1 being not likely at all and 7 being extremely likely)

Avoidance is also measured, as one of the rating scales mentions not saying anything to the victim. Some of the statements are defensive in nature, so depending on how the participants respond, we can determine how defensive they are.

One final measure is accountability. There are several statements in our study that involve the participant rating a statement that takes accountability for their actions.

4. We also measured intellectual humility. We used the general intellectual humility scale (Leary et al., 2017)

Leary, M. R., Diebels, K. J., Davisson, E. K., Jongman-Sereno, K. P., Isherwood, J. C., Raimi, K. T., Deffler, S. A., & Hoyle, R. H. (2017). Cognitive and interpersonal

features of intellectual humility. *Personality and Social Psychology Bulletin*, 43(6), 793-813. <https://doi-org.pitt.idm.oclc.org/10.1177/014616721769769>

5. How did you operationalize ALL demographic variables?

We operationalized all demographic variables by listing out answer choices for each variable, or allowing participants to write their own description.

Example #1

Method

Participants

To power Study 2 to detect the hypothesized serial indirect effects, we conducted a Monte Carlo power analysis (Schoemann et al., 2017) based on our findings from Study 1, which indicated that we would need a sample of 550 to attain 80% power. We recruited 600 participants from Connect on the CloudResearch platform to account for potential loss of participants due to pre-registered data exclusion criteria. We utilized the census-matched option to collect data from a representative sample. Participants were compensated at a rate of \$12/hour. Seven participants met our preregistered data exclusion because they failed an attention check question. Our final sample of 593 participants consisted of 294 women, 297 men, 1 non-binary/third gender, and 1 unspecified ($M_{age} = 44.48$, $SD = 15.29$; $Median_{education} = 4$ -year degree). The distribution of race/ethnicity was 70.66% White, 10.29% Black/African American, 7.25% mixed race/ethnicity, 5.73% Asian and Native Hawaiian/Pacific Islander, 5.56% Latinx, .17% Native American, and .34% other race/ethnicity. A sensitivity analysis on this final sample in G*power (Faul et al., 2007) indicated that we could detect small-medium main effects (independent samples t -test, $d = .23$) with 80% power ($\alpha = .05$), and small-medium interactions (within-between ANOVA) (using the SPSS effect size option $f = .137$, $\eta^2 = .019$) with 80% power ($\alpha = .05$, correlation between repeated measures = .5).

Example #2

Method

Participants

We recruited an international sample of 452 participants from Prolific who were predominantly from the United Kingdom, United States, and Europe (see SM for sample size per

country). Participants were compensated at a rate of \$12/hour. Seventeen participants met our pre-registered data exclusion criteria because they failed an attention check question, leaving a final sample of 435 participants (see Table 1 for demographics). We prescreened to sample participants who identified as Christian ($n = 126$), Jewish ($n = 130$), or Muslim ($n = 138$); 39 participants who indicated their present religion as “other religion or faith” and two participants who did not indicate a religious affiliation were included in analyses of our primary hypotheses but excluded from analyses of moderation by religious affiliation. A sensitivity analysis in G*Power (Faul et al., 2007) indicated that we could detect small-medium associations (Pearsons’s $r = .15$) between our predictors and dependent variables.

Materials and Procedure

Recalled Offenses

Participants were first instructed “to think about something that you have done to hurt, upset, or offend someone. Please think of something that is currently unresolved – for example, something immoral that you did that is unsettled or an ongoing issue or that the person you offended has not yet moved on from.” Participants then entered the initials of the transgressor and described the offense.

Following data collection, an undergraduate research assistant categorized the recalled offenses into the following categories (adapted from Schumann, 2012): criticism (e.g., insults, hurting feelings); deception (e.g., lying; withholding information); betrayal (breaking a promise; infidelity); misunderstanding (e.g., miscommunication; misinterpretation of something said or done); failed obligation (e.g., not submitting a work task; not completing chores); inconvenience (e.g., making extra work for someone; taking up someone’s time); disagreement (e.g., conflict

due to divergent values or opinions); physical (e.g., stepping on someone's foot; pushing someone); possession (e.g., stealing money; breaking a favorite mug).

Offense-specific Divine Forgiveness

Next, participants completed an adapted version of Fincham and May's (2019) 3-item divine forgiveness measure. We adapted the items so that participants indicated their beliefs about divine forgiveness in relation to the offense they just recalled ("I believe that God has forgiven me for the offense," "I believe that God has not yet forgiven me for the offense" [reverse-coded], "I believe that God has released me from the debt of the offense") on a 7-point scale (1 = *Strongly disagree*, 7 = *Strongly agree*; $\alpha = .75$).

Self-Forgiveness

Participants then completed a measure of state self-forgiveness (Wohl et al., 2008). Participants were instructed to answer each item "while contemplating how you feel about yourself 'right now' regarding the unresolved conflict." This measure includes 8 items measuring self-forgiving feelings and actions (e.g., "I feel accepting of myself"), and 9 items measuring self-forgiving beliefs (e.g., "I believe I am a bad person" [reverse-coded]), all answered on a 7-point scale (1 = *Strongly disagree*, 7 = *Strongly agree*; $\alpha = .95$).

Apology Outcomes

Transgressors' Likelihood of Apologizing

Next, participants were asked, "On a scale from 0 (*not at all likely*) to 100 (*Extremely likely*), how likely are you to apologize to [initials] for this offense?"

Apology Email

Participants were then instructed to write an email to the victim, and to use the email to say whatever it is they wanted to say to the victim about the offense (Schumann & Dragotta,

2021). Participants believed they would be sending the email to the victim at the end of the study (but were never asked to send it and were debriefed at the end of the study). Following data collection, the emails were coded by three independent raters, who were blind to participants' scores on all other measures, for the presence of an apology/remorse ($\alpha = .88$). Two different raters, who were blind to participants' scores on all other measures, coded for overall apology quality, $r(401) = .80$, and apology sincerity, $r(401) = .81$. Here and in Study 2, we asked participants whether they apologized in their email (*yes/no*) and whether they had previously apologized for this offense (*yes/no*) (see SM).